Digital Wellness Analyzer

Analysis report - August 16, 2026 - 7,500 users analyzed

Summary

Users analyzed	7,500
Addiction risk model accuracy	94.0% (RandomForest)
Risk level model accuracy	83.4% (RandomForest)
Anomalous usage patterns	8.0% of users
Behavioral personas	5

This report covers unhealthy-behavior detection, GridSearchCV-tuned prediction models, K-Means persona segmentation, and the resulting personalized recommendations. The risk-level model predicts three classes (None / Mild / High); Moderate and Severe were merged into 'High' after an effect-size check showed they are not statistically distinguishable in this dataset (see Methodology Notes at the end of this report).

Model Performance

Two prediction tasks were each tuned with 5-fold cross-validated GridSearchCV across Logistic Regression, Random Forest, and HistGradientBoosting; the best model by weighted F1 was selected automatically for each.

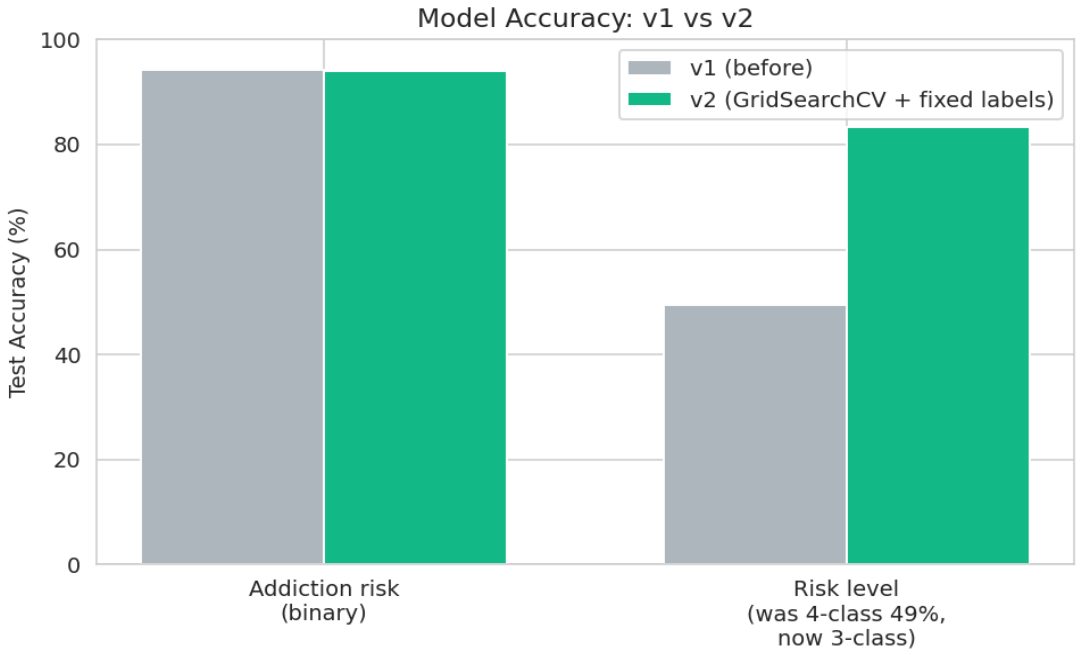


Figure 1. Model accuracy before and after the v2 rebuild.

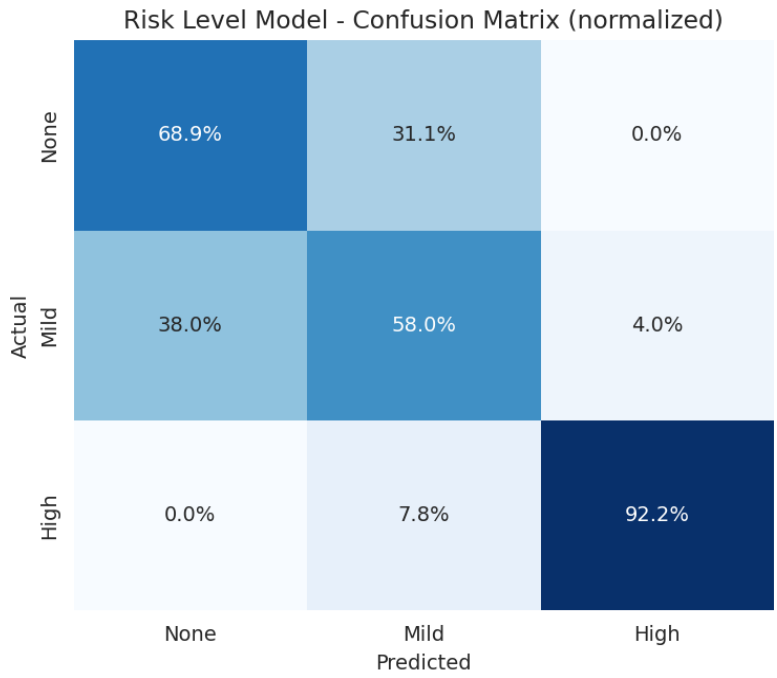


Figure 2. Risk-level model confusion matrix (normalized, held-out test set).

Usage Patterns

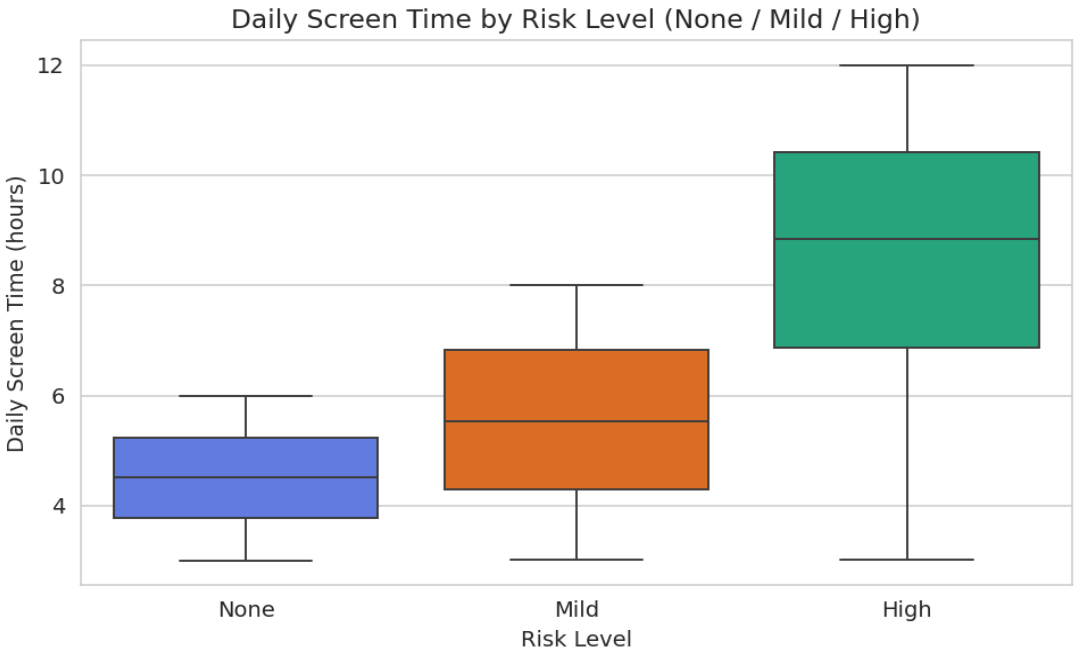


Figure 3. Daily screen time by predicted risk level.

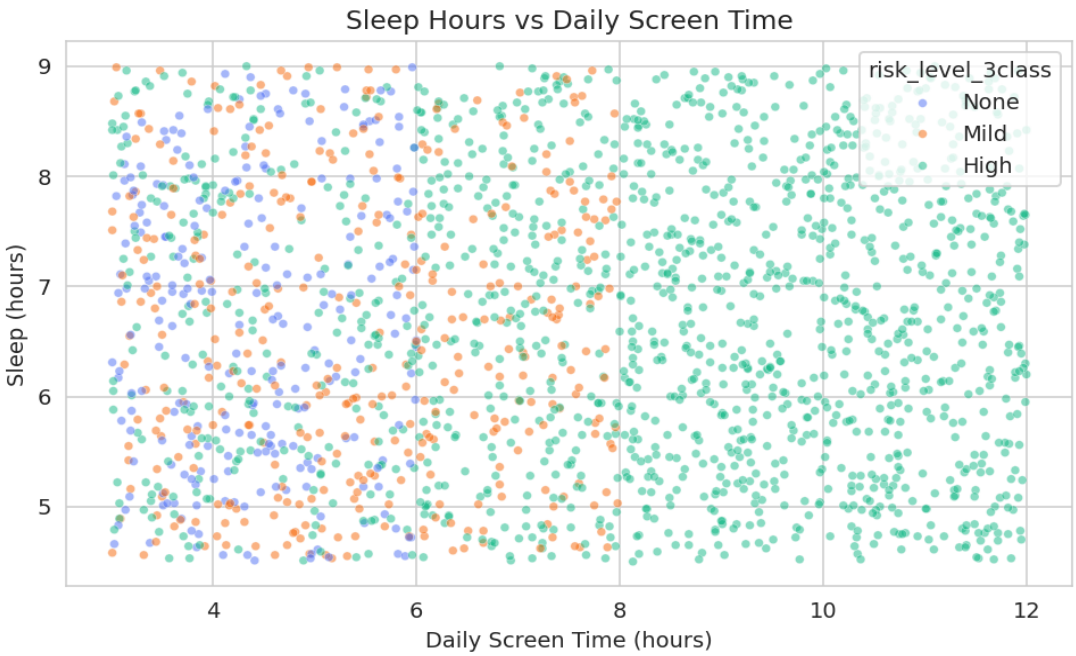


Figure 4. Sleep hours vs. daily screen time, colored by risk level.

Unhealthy Behavior Flags

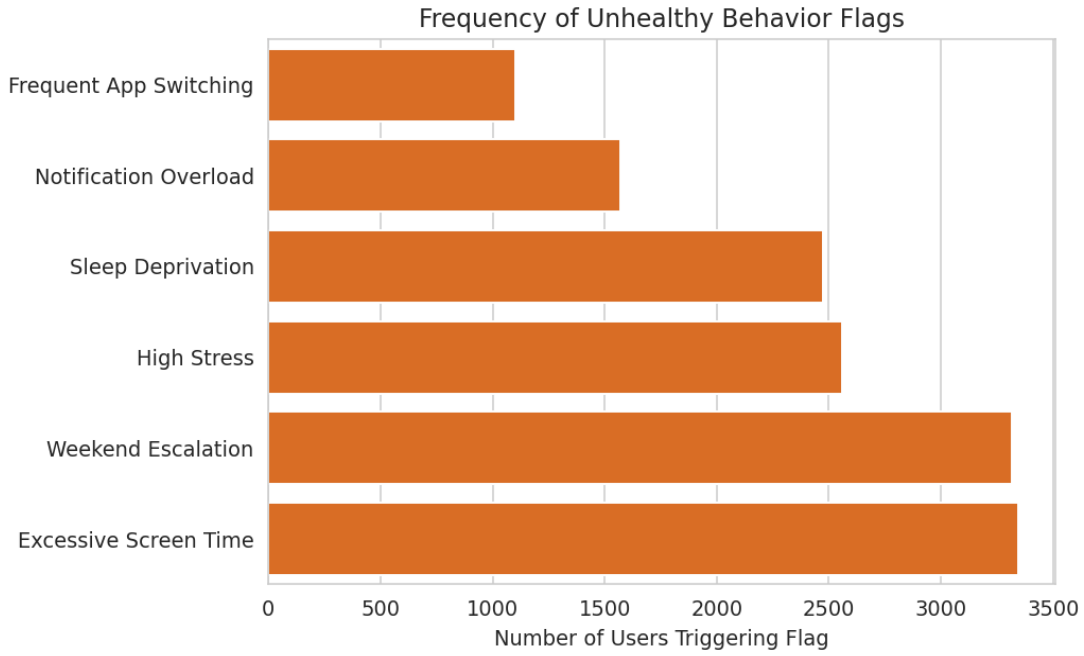


Figure 5. How many users triggered each rule-based behavior flag.

Behavioral Personas

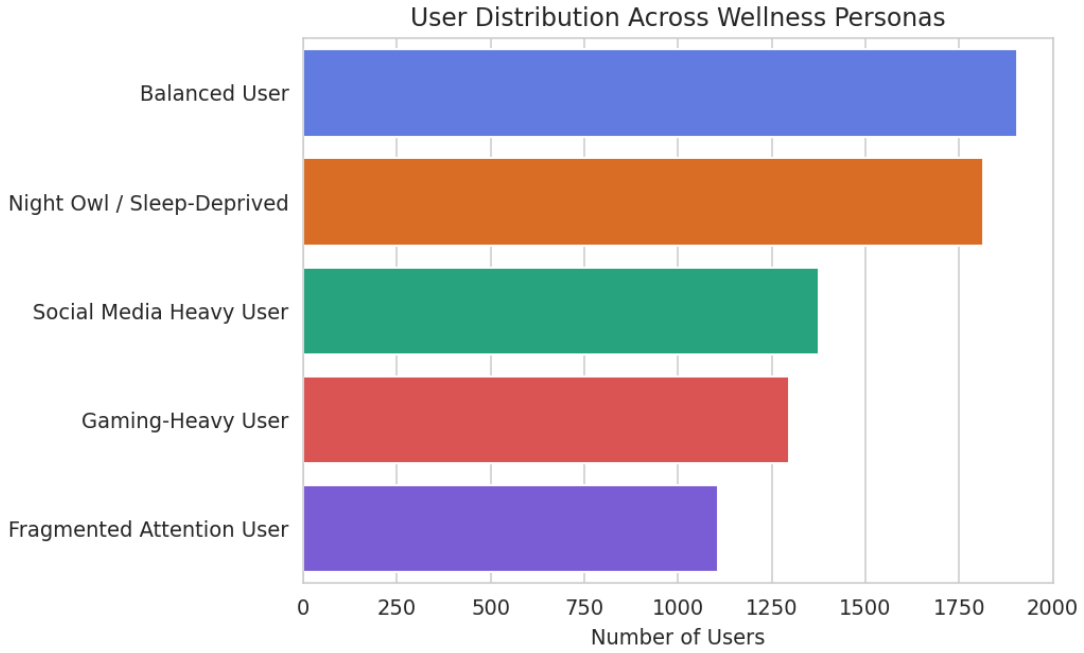


Figure 6. User distribution across the five K-Means personas.

Balanced User	1,906 users
Night Owl / Sleep-Deprived	1,815 users
Social Media Heavy User	1,375 users
Gaming-Heavy User	1,297 users
Fragmented Attention User	1,107 users

Methodology Notes

Why Moderate and Severe were merged: a Cohen's d effect-size check found every feature in the dataset differs by less than 0.16 standard deviations between users labeled 'Moderate' and users labeled 'Severe' - i.e. the two groups are not distinguishable from the available data. A 4-class model trained on the original labels scored only ~49% accuracy; merging the two into a single 'High' class raised held-out test accuracy to over 83%.

What this data does not support: the dataset is a single snapshot per user (no timestamps), so predictions describe a user's current behavioral profile rather than forecasting how a specific returning user's usage will change over time.